Xi'an Jiaotong-Liverpool University

西交利物浦大学

PAPER CODE	EXAMINER	DEPARTMENT	TEL
CPT103		Computing	

2nd SEMESTER 2024/25 FINAL EXAMINATION

Undergraduate – Year 2

Introduction to Database Systems

TIME ALLOWED: 2 Hours

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INSTRUCTIONS TO CANDIDATES

1、 This is a closed book examination.

2、 Total marks available are 100. This will count for 70% in the final assessment.

3、 Answer all questions.

4、 Answer should be written in the answer booklet(s) and MCQ card(s) provided.

5、 Only English solutions are accepted.

6、 All materials must be returned to the exam supervisor upon completion of the exam. Failure to do so will be deemed academic misconduct and will be dealt with accordingly.

Question A（40 marks）

Multiple-Choice

1. A primary key of a relation must be:
   A. Always numeric
   B. Unique and NOT NULL
   C. A single attribute
   D. The first column in the table
2. A candidate key is:
   A. Any foreign key
   B. Any attribute that is NOT NULL
   C. A minimal set of attributes that uniquely identifies tuples
   D. A key chosen randomly
3. A foreign key is:
   A. A copy of a key in the same table
   B. A column that references a key in another table
   C. A column that must be unique
   D. A column that cannot contain NULL
4. Referential integrity ensures:
   A. All attributes are atomic
   B. Foreign key values match existing parent key values
   C. All NULLs are removed
   D. Every table has a default value
5. If a foreign key uses ON DELETE CASCADE:
   A. Deleting a parent row is forbidden
   B. Child foreign keys become NULL
   C. Matching child rows are deleted
   D. Nothing happens to child rows

6. Consider table Employee(Name, Salary), where one row has Salary = NULL.
   What does the following query return regarding the row with Salary = NULL?

```
SELECT * FROM Employee
WHERE Salary > 15000;
```

   A. The row is always included.
   B. The row is included only if NULL is treated as 0.
   C. The row is excluded because the condition evaluates to **unknown**.
   D. The row is included and NULL is replaced by the average salary.

7. For the condition WHERE salary > 3000:
   A. NULL salaries always satisfy the condition
   B. NULL salaries never satisfy the condition
   C. NULL is treated as 0
   D. NULL is replaced by 3000
8. In aggregate functions:
   A. NULL values are treated as 0
   B. NULL values are ignored
   C. NULL values cause errors
   D. NULL values become defaults

9. Which statement correctly **adds** a new column email (up to 50 characters, may be NULL) to table staff?

   A. ALTER TABLE staff

      INSERT COLUMN email VARCHAR(50);

   B. ALTER TABLE staff

      ADD email VARCHAR(50);

   C. ALTER TABLE staff

      ADD COLUMN email CHAR(50) NOT NULL;

   D. ALTER TABLE staff

      NEW COLUMN email VARCHAR(50);

10. A LEFT OUTER JOIN returns:
    A. Only matching rows
    B. All rows from the right table
    C. All rows from the left table with NULLs for missing matches
    D. Only non-matching rows
11. GROUP BY is used to:
    A. Sort rows
    B. Filter rows before aggregation
    C. Group rows for aggregate functions
    D. Remove duplicate rows only

12. Consider the query:

```
SELECT ID, Name, aStreet, aTown, aPostcode, pNumber, Classification
FROM ((Student LEFT OUTER JOIN Phone
            ON Student.pID = Phone.pID)
        LEFT OUTER JOIN Address
            ON Student.aID = Address.aID)
        INNER JOIN Degree
            ON Student.ID = Degree.ID
WHERE Grad = 'A';
```

What is the main **benefit** of using LEFT OUTER JOINs here?

A. To exclude students without phone or address records.
B. To ensure that students with missing phone or address data still appear in the result.
C. To speed up the query using indexes.
D. To automatically fill missing phone numbers with default values.

13. Two queries can be combined with UNION if:
    A. They use the same table
    B. They have the same number of columns with compatible types
    C. Both use DISTINCT
    D. Both use GROUP BY
14. A functional dependency A → B means:
    A. B determines A
    B. A and B have identical values
    C. Each value of A is associated with exactly one value of B
    D. A must be a primary key
15. A partial dependency occurs when:
    A. A non-key attribute depends on the whole key
    B. A non-key attribute depends on part of a composite key
    C. A key attribute depends on a non-key attribute
    D. Two attributes are unrelated
16. A relation is in 2NF if:
    A. It has no transitive dependencies
    B. It has no repeating groups
    C. It is in 1NF and has no partial dependencies
    D. It has exactly one candidate key

17. A relation is in 3NF if:
    A. Every determinant is a candidate key
    B. Non-key attributes do not depend transitively on candidate keys
    C. There is only one non-key attribute
    D. It contains no foreign keys

18. Consider the foreign key:

```
CONSTRAINT FK_staff_branchNo
FOREIGN KEY (branchNo)
REFERENCES branch (branchNo)
ON DELETE SET NULL
ON UPDATE CASCADE;
```

What happens if a row in branch with branchNo = 'B005' is **updated to 'B010'**?

A. The update is rejected (RESTRICT).
B. All referencing staff.branchNo = 'B005' become NULL.
C. All referencing staff.branchNo = 'B005' become 'B010'.
D. All rows in staff are deleted.

19. The main goal of normalization is to:
    A. Speed up all SQL queries
    B. Reduce redundancy and anomalies
    C. Remove the need for joins
    D. Automatically add indexes
20. The statement SELECT name FROM Student; does:
    A. Updates the name column
    B. Deletes rows
    C. Retrieves all name values from the table
    D. Creates a new table

Question B (20 marks)

1) Explain what is JDBC.

(4 marks)

2) Explain what is the atomicity and durability properties of transactions.

(6 marks)

3) Explain the purpose of data encryption in database in terms of preserving user privacy.

(4 marks)

4) Explain the purpose of a CHECK constraint and give an example of it inside a CREATE TABLE statement.

(6 marks)

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Question C (15 marks)

Normalise the following table *T* into the 3rd normal form by clearly describing the normalisation process, i.e., the dependencies removed and how the table is split into sub-tables. In each normalisation step, clearly describe the primary keys and functional dependencies of each resulting sub-table.

A	B	C	D	E	F	G	H

Attributes (C, D, E) form the primary key. Other functional dependencies are listed below:

```
C, D -> F, G, H

F -> D

F, G -> H

E -> B
```

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Question D (25 marks)

You are asked to develop a database system for a cinema to streamline its movie management operations . The requirements for the database are provided below:

1. Each movie should have a unique ID, name, the release year and duration.
2. Each movie can have one or more actors playing in it, and the system should record actors' basic demographic information with movies they have played.
3. The system should maintain professional profiles for directors, including demographic and operational identifiers. It should also keep track of movies that each director has directed.
4. A Movies can belong to one or more genres (e.g., "Action", "Comedy").
5. Each movie can be produced from only one Studio (e.g. "Disney", "Warner Bros. ").
6. Each user should have an account in the system registered using their email address and can write reviews for movies, including a rating (1 – 10) and a comment.

Draw the entity relationship diagram. Make sure you underline primary keys in the diagram. No M:N or 1:1 relationships are allowed in your answer.

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END OF FINAL EXAM